

VIKRAM SHIVAKUMAR

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EDUCATION

Johns Hopkins University PhD Candidate, Computer Science Advisor: Ben Langmead	2021-
Johns Hopkins University M.S. in Biomedical Engineering	2023
University of California, Berkeley B.S. in Bioengineering (Honors) Minor in Computer Science	2020

PROFESSIONAL EXPERIENCE

Undergraduate Researcher UC Berkeley, Department of Bioengineering Berkeley, CA Advisor: Liana Lareau	2021
Research Intern Georgetown University, Innovation Center for Biomedical Informatics (ICBI) Washington D.C. Advisor: Matthew McCoy	2018-2019
Research Intern National Museum of Natural History, Lab of Analytical Biology (LAB) Washington D.C. Advisor: Elizabeth Zimmer	2015-2017

FELLOWSHIPS AND AWARDS

National Science Foundation Graduate Research Fellowship (Total amount: \$138,000)	2021-2026
Best Short Talk and Poster — RECOMB-seq (Istanbul, Turkey)	2023
Best Proceedings Paper and Talk — ISMB (Montreal, QC)	2024
Best Poster (2 nd place) — ARCH HPC Symposium (JHU)	2025

PUBLICATIONS

Research Articles

8. S. Kovaka, P.W. Hook, K.M. Jenike, **V. Shivakumar**, L.B. Morina, R. Razaghi, W. Timp, M.C. Schatz (2025). Uncalled4 improves nanopore DNA and RNA modification detection via fast and accurate signal alignment. *Nature Methods*, 1–11.
7. N.K. Brown, **V.S. Shivakumar**, B. Langmead (2025). Improved pangenomic classification accuracy with chain statistics. *International Conference on Research in Computational Molecular Biology (RECOMB)*, 190–208.

6. **V.S. Shivakumar**, O.Y. Ahmed, S. Kovaka, M. Zakeri, B. Langmead (2024). Sigmoni: classification of nanopore signal with a compressed pangenome index. *Bioinformatics*, 40(Supplement_1), i287–i296.
5. A.L. Cope, F. Anderson, J. Favate, M. Jackson, A. Mok, A. Kurowska, J. Liu, E. MacKenzie, **V. Shivakumar**, P. Tilton, S.M. Winterbourne, S. Xue, K. Kavoussanakis, L.F. Lareau, P. Shah, E.W.J. Wallace (2022). riboviz 2: a flexible and robust ribosome profiling data analysis and visualization workflow. *Bioinformatics*, 38(8), 2358–2360.
4. M.D. McCoy, **V. Shivakumar**, S. Nimmagadda, M.S. Jafri, S. Madhavan (2019). SNP2SIM: a modular workflow for standardizing molecular simulation and functional analysis of protein variants. *BMC Bioinformatics*, 20, 1–8.
3. G. Damasco, **V.S. Shivakumar**, T.M. Misciewicz, D.C. Daly, P.V.A. Fine (2019). Leaf transcriptome assembly of *Protium copal* (Burseraceae) and annotation of terpene biosynthetic genes. *Genes*, 10(5), 392.
2. **V.S. Shivakumar**, G. Johnson, E.A. Zimmer (2019). Transcriptome analysis of the curry tree (*Berbera koenigii* L., Rutaceae) during leaf development. *Scientific Reports*, 9(1), 4230.
1. **V.S. Shivakumar**, M.S. Appelhans, G. Johnson, M. Carlsen, E.A. Zimmer (2017). Analysis of whole chloroplast genomes from the genera of the Clauseneae, the curry tribe (Rutaceae, Citrus family). *Molecular Phylogenetics and Evolution*, 117, 135–140.

Preprints

3. **V.S. Shivakumar**, B. Langmead (2025). Mumemto: efficient maximal matching across pangenomes. *bioRxiv*. (In review at Genome Biology)
2. A. Zheng, I. Lee, **V.S. Shivakumar**, O.Y. Ahmed, B. Langmead (2025). Fast and flexible minimizer digestion with digest. *bioRxiv*. (In review at Bioinformatics)
1. **V. Shivakumar**, W. Reid, S. Madhavan, M.D. McCoy (2020). Simulating Protein Structure to Support Clinical Decisions: Predicting the Ligand Specific Impact of Genomic Variation on Opioid Binding. *Research Square*.

PRESENTATIONS

Invited Talks

6. **Mumemto: efficient maximal matching across pangenomes**
Wellcome Sanger Institute seminar series, Hinxton, UK. July 30, 2025.
5. **Mumemto: efficient maximal matching across pangenomes**
Human Pangenome Reference Consortium (HPRC) meeting, presented remotely. March 24, 2025.
4. **Sigmoni: efficient pangenome multi-classification of nanopore signal**
Intelligent Systems for Molecular Biology (ISMB). Montreal, Canada. July 13, 2024.
3. **Mumemto: efficient maximal matching across multiple genomes**
RECOMB-seq, Cambridge, MA. April 27, 2024.
2. **Sigmoni: efficient pangenome multi-classification of nanopore signal**
Workshop on efficient methods for sequence analysis (WEMSA). State College, PA. May 24, 2023.

1. **Sigmoni: efficient pangenome multi-classification of nanopore signal**
Research on Computational Molecular Biology: Satellite Conference on Biological Sequence Analysis (RECOMB-seq). Istanbul, Turkey. April 14, 2023.

Poster Presentations

3. **Mumemto: efficient maximal matching across pangenomes**
 - Intelligent Systems for Molecular Biology (ISMB). Liverpool, UK. July 21-24, 2025.
 - Biology of Genomes. Cold Spring Harbor Laboratory, NY, USA. May 9, 2025.
 - High Performance Computing Symposium, Johns Hopkins University. Baltimore, MD. April 2, 2025.
 - Workshop on efficient methods for sequence analysis (WEMSA). State College, PA. August 9, 2024.
 - RECOMB-seq, Cambridge, MA. April 27, 2024.
2. **Sigmoni: efficient pangenome multi-classification of nanopore signal**
 - Genome Informatics. Cold Spring Harbor Laboratory, NY, USA. December 8, 2023.
 - Workshop on efficient methods for sequence analysis (WEMSA). State College, PA. May 24, 2023.
 - Research on Computational Molecular Biology (RECOMB). Istanbul, Turkey. April 16, 2023.
 - Biological Data Science. Cold Spring Harbor Laboratory, NY, USA. November 10, 2022.
1. **Analysis of whole Chloroplast genomes from the genera of the Clauseneae, the Curry tribe (Rutaceae, Citrus family)** Boyany. Savannah, GA. August 1st, 2016.

TEACHING AND MENTORSHIP

Teaching

2020 TA – BioE/CompBio 131, Introduction to Computational Biology (Prof. Ian Holmes)
2021 CA – BioE/CompBio 145, Introduction to Computational Biology (Prof. Liana Lareau)
2023 TA (and guest lecturer) – Sketching and Indexing for Sequences (Prof. Ben Langmead)
2024 CA (and guest lecturer) – Sketching and Indexing for Sequences (Prof. Ben Langmead)

Mentorship

Students supervised:

- Alan Zheng, undergraduate researcher
- Ishmael Lee, undergraduate researcher
- Andrew Raptis, high school intern
- Arjun Garg, high school intern

SOFTWARE

Mumemto - Exact matching across pangenomes

<https://github.com/vikshiv/mumemto>

Sigmoni - fast Nanopore signal classification using compressed indexing

<https://github.com/vikshiv/sigmoni>

Digest - C++ and Python API for minimizer digestion schemes

<https://github.com/VeryAmazed/digest>

riboviz - visualizing ribosome profiling sequencing data

<https://github.com/riboviz/riboviz>

SNP2SIM - variant-based molecular dynamics

<https://github.com/mccoymd/snp2sim>

SERVICE AND MEMBERSHIPS

Peer Review Experience

2025 – Bioinformatics (reviewer)

2023 – RECOMB (sub-reviewer)

2023 – ISMB (sub-reviewer)

Memberships

2023 – International Society for Computational Biology (ISCB)

2025 – Human Pangenome Reference Consortium (HPRC)

Service

2021 Volunteer - Orientation, Biomedical Engineering department

2018 – 2020 Treasurer of UC Berkeley Molecular and Cell Biology Student Organisation (MCBcDNA)